

THE FUNDAMENTAL LEMMA OF GAME PLAYING

ADVANCED TOPICS IN ~~CYBERSECURITY~~ CRYPTOGRAPHY (7CCSMATC)

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OUTLINE

Introduction

CTR Mode

Fundamental Lemma of Game Playing

Proof of the Fundamental Lemma of Game Playing

Closing

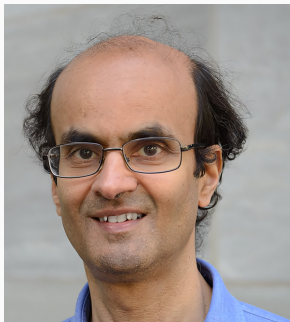
INTRODUCTION

- We have defined what it means for an encryption scheme to be secure (IND-CPA + INT-CTXT = IND-CCA).
- We have shown that the OTP achieves IND-CPA security, even **unconditionally**.

The One-Time Pad is impractical, we want something more manageable $\Rightarrow$
Pseudorandomness!

MAIN REFERENCE

Mihir Bellare and Phillip Rogaway. **Code-Based Game-Playing Proofs and the Security of Triple Encryption**. Cryptology ePrint Archive, Report 2004/331. 2004. URL: <https://eprint.iacr.org/2004/331>

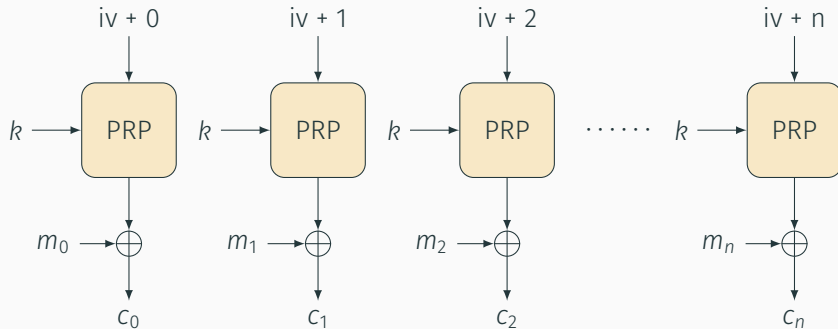


Mihir Bellare is a professor at UCSD

- 2003** RSA Conference's Sixth Annual Award
- 2013** Fellow of the Association for Computing Machinery.
- 2019** Levchin Prize for Real-World Cryptography

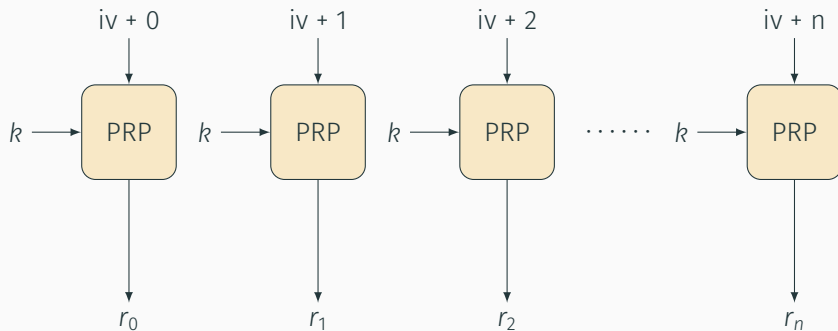
CTR MODE

CTR MODE



Picture credit: <https://www.iacr.org/authors/tikz/>

CTR MODE STREAM



$$r_i \in \{0, 1\}^\lambda$$

WANT: $n + 1$ PSEUDORANDOM STRINGS OF LENGTH λ

Definition (PRF)

A PRF is a keyed function $F_k : \{0, 1\}^\lambda \rightarrow \{0, 1\}^\ell$ where ℓ depends on λ and $k \leftarrow \mathcal{K}$. We say F_k is (t, ε) -secure **PRF** if for Game₀ and Game₁ defined below we have:

$$\forall \mathcal{D} \in t \text{ steps: } \text{Adv}_F^{\text{prf}}(\mathcal{D}) = |\Pr[\mathcal{D}^{\text{Game}_1} = 1] - \Pr[\mathcal{D}^{\text{Game}_0} = 1]| < \varepsilon$$

Game₀

1: $f \leftarrow \emptyset$

2: **return** $\mathcal{D}^F$

Game₁

1: $f \leftarrow \emptyset; k \leftarrow \mathcal{K}$

2: **return** $\mathcal{D}^F$

$F(x)$

1: **if** $x \notin f.\text{keys}$ **then** $f[x] \leftarrow \{0, 1\}^\ell$

2: $y \leftarrow f[x]$

3: $y \leftarrow F_k(x)$ //Game₁

4: **return** y

HAVE: $n + 1$ CALLS TO PSEUDORANDOM PERMUTATION OF LENGTH λ

Definition (PRP)

A PRP is a keyed permutation $E_k : \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ for $k \leftarrow \mathcal{K}$. We say E is (t, ε) -secure PRP if for Game_0 and Game_1 defined below we have:

$$\forall \mathcal{D} \in t \text{ steps: } \text{Adv}_E^{\text{prp}}(\mathcal{D}) = |\Pr[\mathcal{D}^{\text{Game}_1} = 1] - \Pr[\mathcal{D}^{\text{Game}_0} = 1]| < \varepsilon$$

Game ₀	P(x)
1: $f \leftarrow \emptyset$	1: if $x \notin f.\text{keys}$ then $f[x] \leftarrow \{0, 1\}^\lambda \setminus f.\text{values}$
2: return $\mathcal{D}^P$	2: $y \leftarrow f[x]$
Game ₁	3: $y \leftarrow E_k(x)$ //Game ₁
1: $f \leftarrow \emptyset; k \leftarrow \mathcal{K}$	4: return y
2: return $\mathcal{D}^P$	

Game ₀	F(x)
1: $f \leftarrow \emptyset$	1: if $x \in f.\text{keys}$ then
2: return $\mathcal{D}^F$	2: $y \leftarrow f[x]$
Game ₁	3: else
1: $f \leftarrow \emptyset;$	4: $y \leftarrow \$ \{0, 1\}^\lambda$
2: return $\mathcal{D}^F$	5: $y \leftarrow \$ \{0, 1\}^\lambda \setminus f.\text{values}$ //Game ₁
	6: $f[x] \leftarrow y$
	7: return y

PRP-PRF SWITCHING LEMMA

Lemma

Let π be a random **permutation** from $\{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$; let ρ be a random **function** from $\{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$. Let $\mathcal{A}$ be an adversary making at most q queries to its oracle, then:

$$|\Pr[\mathcal{A}^\pi] - \Pr[\mathcal{A}^\rho]| \leq \frac{q \cdot (q - 1)}{2^{\lambda+1}}.$$

PRP-PRF SWITCHING LEMMA I

Consider the following games:

Game ₀	P(x)
1: $\pi \leftarrow \emptyset$	1: if $x \in \pi.\text{keys}$ then return $\pi[x]$
2: return $\mathcal{A}^P$	2: $y \leftarrow_{\$} \{0,1\}^\lambda$
Game ₁	3: if $y \in \pi.\text{values}$ then
1: $\pi \leftarrow \emptyset$	4: $\text{bad} \leftarrow \text{true}$
2: return $\mathcal{A}^P$	5: $y \leftarrow_{\$} \{0,1\}^\lambda \setminus \pi.\text{values}$ // Game ₁
	6: $\pi[x] \leftarrow y$
	7: return y

$$|\Pr[\mathcal{A}^\pi] - \Pr[\mathcal{A}^\rho]| = |\Pr[\mathcal{A}^{\text{Game}_0}] - \Pr[\mathcal{A}^{\text{Game}_1}]| \quad (1)$$

$$\leq \Pr[\mathcal{A}^{\text{Game}_0}] \text{ sets bad} \quad (2)$$

$$\leq q \cdot (q - 1) / 2^{\lambda+1} \quad (3)$$

On Eq. (1): Game₀ perfectly simulates a random function ρ and Game₁ perfectly simulates a random permutation π , by the **principle of lazy sampling**.

Thus, we have

$$\Pr[\mathcal{A}^\rho] = \Pr[\mathcal{A}^{\text{Game}_0}] \text{ and } \Pr[\mathcal{A}^{\text{Game}_1}] = \Pr[\mathcal{A}^\pi].$$

On Eq. (2): we will appeal to the **fundamental lemma of game playing**.

On Eq. (3): by the union bound the probability that $y \in \pi.\text{values}$, is at most

$$\frac{(1 + 2 + \cdots + (q - 1))}{2^\lambda} = \frac{q \cdot (q - 1)}{2^{\lambda+1}}.$$

FUNDAMENTAL LEMMA OF GAME PLAYING

We say Game_0 and Game_1 are “identical-until-bad” if they are ... identical until some flag bad is set.

FUNDAMENTAL LEMMA OF GAME PLAYING

Lemma (Fundamental Lemma of Game Playing)

Let Game_0 , Game_1 be identical-until-bad games and $\mathcal{A}$ be an adversary. Then

$$|\Pr[\mathcal{A}^{\text{Game}_0}] - \Pr[\mathcal{A}^{\text{Game}_1}]| \leq \Pr[\mathcal{A}^{\text{Game}_0} \text{ sets bad}].$$

PROOF OF THE FUNDAMENTAL LEMMA OF GAME PLAYING

We follow

Mike Rosulek. **The Joy of Cryptography**. <https://joyofcryptography.com>. self published, 2021, Proof of Lemma 4.8

CONDITIONAL PROBABILITIES

- If Y is some event, then we write $\bar{Y}$ to denote the complement event i.e the event that Y does not happen. For all events Y , we have

$$\Pr[Y] + \Pr[\bar{Y}] = 1$$

- We write $\Pr[X | Y]$ to denote the probability of X , conditioned on the event Y . Conditional probability is defined as:

$$\Pr[X | Y] = \Pr[X \wedge Y] / \Pr[Y]$$

- It satisfies the important identity for all events X and Y :

$$\Pr[X] = \Pr[X | Y] \cdot \Pr[Y] + \Pr[X | \bar{Y}] \cdot \Pr[\bar{Y}].$$

- Let B_0 be the event that bad is set in Game₀
- Let B_1 be the event that bad is set in Game₁
- We can write:

$$\Pr[\mathcal{A}^{\text{Game}_0}] = \Pr[\mathcal{A}^{\text{Game}_0} \mid B_0] \cdot \Pr[B_0] + \Pr[\mathcal{A}^{\text{Game}_0} \mid \bar{B}_0] \cdot \Pr[\bar{B}_0]$$

$$\Pr[\mathcal{A}^{\text{Game}_1}] = \Pr[\mathcal{A}^{\text{Game}_1} \mid B_1] \cdot \Pr[B_1] + \Pr[\mathcal{A}^{\text{Game}_1} \mid \bar{B}_1] \cdot \Pr[\bar{B}_1]$$

- We have $\alpha := \Pr[B_0] = \Pr[B_1]$ because the two games are identical-until-bad.
- We have $\beta := \Pr[\mathcal{A}^{\text{Game}_0} \mid \bar{B}_0] = \Pr[\mathcal{A}^{\text{Game}_1} \mid \bar{B}_1]$ because the two games only differ when bad is set.

- Substituting in, we obtain

$$\Pr[\mathcal{A}^{\text{Game}_0}] = \Pr[\mathcal{A}^{\text{Game}_0} \mid B_0] \cdot \alpha + \beta \cdot (1 - \alpha)$$

$$\Pr[\mathcal{A}^{\text{Game}_1}] = \Pr[\mathcal{A}^{\text{Game}_1} \mid B_1] \cdot \alpha + \beta \cdot (1 - \alpha)$$

- We can calculate the advantage

$$\begin{aligned} |\Pr[\mathcal{A}^{\text{Game}_0}] - \Pr[\mathcal{A}^{\text{Game}_1}]| &= \left| \Pr[\mathcal{A}^{\text{Game}_0} \mid B_0] \cdot \alpha + \beta \cdot (1 - \alpha) - (\Pr[\mathcal{A}^{\text{Game}_1} \mid B_1] \cdot \alpha + \beta \cdot (1 - \alpha)) \right| \\ &= |(\Pr[\mathcal{A}^{\text{Game}_0} \mid B_0] - \Pr[\mathcal{A}^{\text{Game}_1} \mid B_1]) \cdot \alpha| \\ &= |\Pr[\mathcal{A}^{\text{Game}_0} \mid B_0] - \Pr[\mathcal{A}^{\text{Game}_1} \mid B_1]| \cdot \alpha \\ &\leq \alpha = \Pr[B_0] = \Pr[\mathcal{A}^{\text{Game}_0} \text{ sets bad}]. \end{aligned}$$

where the last step follows from $0 \leq |\Pr[\mathcal{A}^{\text{Game}_0} \mid B_0] - \Pr[\mathcal{A}^{\text{Game}_1} \mid B_1]| \leq 1$.

CLOSING

MATCHING ATTACK

- Call $\sqrt{2^\lambda} = 2^{\lambda/2}$ times and check if any answer repeats.
- By the birthday bound this happens with constant probability

Memory-less Attack

Read about the Pollard-rho attack to learn how to make this attack use $\text{poly}(\lambda)$ memory instead of $2^{\lambda/2}$.

FIN

WE WANT TO APPROXIMATE THE ONE-TIME PAD

IF WE HAVE A PRF, THIS IS STRAIGHT-FORWARD

IF WE “ONLY” HAVE A PRP, **AN IDEAL PRIMITIVE**, THIS BREAKS
DOWN AFTER $q = \sqrt{2^\lambda}$ QUERIES, E.G. 2^{64} FOR $\lambda = 128$
(AES-128).

NEXT: HOW DO WE GET A PRP?

- [BR04] Mihir Bellare and Phillip Rogaway. **Code-Based Game-Playing Proofs and the Security of Triple Encryption**. Cryptology ePrint Archive, Report 2004/331. 2004. URL: <https://eprint.iacr.org/2004/331>.
- [Ros21] Mike Rosulek. **The Joy of Cryptography**. <https://joyofcryptography.com>. self published, 2021.

We require that both the adversary and the game always terminate in finite time.

- For any adversary $\mathcal{A}$ there must exist an integer T such that $\mathcal{A}$ always halts within T steps (regardless of the random choices $\mathcal{A}$ makes and the answers it receives to its oracle queries).
- For any game Game there must exist an integer T such that Game always halts within T steps (regardless of the random choices made).

Since $\mathcal{A}$ and Game terminate in finite time,

- there must be an integer T such that they each execute at most T random-assignment statements, and
- there must be an integer B such that the size of the set $\mathcal{S}$ in any random-assignment statement $s \leftarrow \$ \mathcal{S}$ executed by the adversary or the game is at most B .

$\Rightarrow$ The execution of Game with $\mathcal{A}$ uses finite randomness, meaning Game and $\mathcal{A}$ are underlain by a finite sample space Ω .

Punchline

Probabilities are well-defined and we can talk about the probabilities of various events in the execution.

- This means that there exists an integer z such that the execution of Game_0 with $\mathcal{A}$ and the execution of Game_1 with $\mathcal{A}$ perform no more than z random-assignment statements, each of these sampling from a set of size at most z .

PROOF OF THE FUNDAMENTAL LEMMA OF GAME PLAYING AS IN [BR04] IV

- Let $\mathcal{C} := \text{Coins}(\mathcal{A}, \text{Game}_0, \text{Game}_1) = [1 \dots z!]^z$ be the set of z -tuples of numbers, each number between 1 and $z!$.

```
z = 2
R = IntegerModRing(factorial(z)); offset = vector(R, z, [1]*z).lift()
Coins = [coin.lift() + offset for coin in FreeModule(R, z)]
print(Coins)
```

- For $\mathbf{c} = (c_0, \dots, c_{z-1}) \in \mathcal{C}$, the execution of Game with $\mathcal{A}$ on coins $\mathbf{c}$ is defined as follows:
 - On the i -th random-assignment statement, call it $x \leftarrow \mathcal{U}(\mathcal{S})$, where $\mathcal{S} := \{s_i\}_{0 \leq i < m}$, if $\mathcal{S} \neq \emptyset$, return $s_{c_i \bmod |\mathcal{S}|}$, otherwise return $\perp$.
- This way to perform random-assignment statements is done regardless of whether it is $\mathcal{A}$ or one of the procedures from Game that is performing the random-assignment statement.

- Note that $m = |\mathcal{S}|$ satisfies $m|z|$ so if $\mathbf{c}$ is chosen at random from $\mathcal{C}$ then the mechanism above will return a point x drawn uniformly from $\mathcal{S}$, and also the values for each random-assignment statement are independent.

- For $\mathbf{c} \in \mathcal{C}$ we let $\text{Game}_0^{\mathcal{A}}(\mathbf{c})$ denote the output of Game_0 when Game_0 is executed with $\mathcal{A}$ on coins $\mathbf{c}$. Same for Game_1 .
- Write $\mathcal{C}_{i,\text{one}} := \{\mathbf{c} \in \mathcal{C} : \text{Game}_i^{\mathcal{A}}(\mathbf{c}) \Rightarrow 1\}$
- Write $\mathcal{C}_i^{\text{bad}} \subseteq \mathcal{C}$ for the coins that result in bad being set to **true** when running $\text{Game}_i^{\mathcal{A}}$.
- Partition $\mathcal{C}_{i,\text{one}}$ into $\mathcal{C}_{i,\text{one}}^{\text{bad}}$ and $\mathcal{C}_{i,\text{one}}^{\text{good}}$ depending on whether bad was set or not in game Game_i .
- Because games Game_0 and Game_1 are identical-until-bad, an element $\mathbf{c} \in \mathcal{C}$ is in $\mathcal{C}_{0,\text{one}}^{\text{good}}$ if and only if it is in $\mathcal{C}_{1,\text{one}}^{\text{good}}$.
 - Bad is never set so the sets are same and in particular have the same size.

We then get:

$$\begin{aligned}\Pr[\text{Game}_0^{\mathcal{A}}] - \Pr[\text{Game}_1^{\mathcal{A}}] &= \frac{c_{0,\text{one}}}{c} - \frac{c_{1,\text{one}}}{c} \\ &= \frac{c_{0,\text{one}}^{\text{good}} + c_{0,\text{one}}^{\text{bad}}}{c} - \frac{c_{1,\text{one}}^{\text{good}} + c_{1,\text{one}}^{\text{bad}}}{c} \\ &= \frac{c_{0,\text{one}}^{\text{bad}}}{c} - \frac{c_{1,\text{one}}^{\text{bad}}}{c} \\ &\leq \frac{c_{0,\text{one}}^{\text{bad}}}{c} \\ &\leq \frac{c_0^{\text{bad}}}{c} \\ &= \Pr[\text{Game}_0^{\mathcal{A}} \text{ sets bad}].\end{aligned}$$